Prediction Algorithm for Search through miRNA Databases

Alja Videtič Paska[1], Iris Šalamon[1], Mojca Katrašnik[1], Urban Alič[2]; Tomaž Zupanc[3], Katarina Kouter[1]

[1] Institute of Biochemistry and Molecular Genetics, Faculty of Medicine, University of Ljubljana, Vrazov trg 2, 1000 Ljubljana, Slovenia
[2] Faculty of Medicine, University of Ljubljana, Vrazov trg 2, 1000 Ljubljana, Slovenia
[3] Institute of Forensic Medicine, Faculty of Medicine, University of Ljubljana, Ljubljana, Slovenia

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Micro RNAs (**miRNAs**) are short noncoding RNAs and there have been more than **2500 different miRNAs identified in humans**. MiRNAs function as post-transcriptional gene regulators at the mRNA level, and represent important epigenetic mechanism of gene downregulation. MiRNAs are of specific interest in study of complex diseases, such as psychiatric disorders, as neuronal miRNA represent 70% of the total body. Since one mRNA molecule can be regulated through several miRNAs and one miRNA can regulate several different mRNAs, is identification of potential miRNAs a quite elaborate task, but the use of modern ***in silico*** approaches could facilitate the work greatly.

The probability and strength of interaction between miRNA and mRNA molecules depends primarily on the nucleotide sequence of both molecules. Several different databases/web algorithms, such as DIANA microT, miRDB, miRmap, miRWalk, and TargetScan, which collect and store data on miRNA are available. These databases use different algorithms and therefore the annotation of miRNA–mRNA pairs can vary.

To overcome limitations of candidate miRNA and whole genome sequencing approaches, we created an ***in silico analysis algorithm*** that would help select the best suitable miRNAs that target the most interesting genes associated with particular brain disease. The **algorithm** enables the **evaluation of robustness of the results of particular miRNAs targeting and their expression in the brain** from several distinct data sources (databases). The sorting tool for miRNA target prediction analysis – **miRNAtpa** – is available at https://github.com/ualich/miRNAtpa (Figure 1) (1). The analysis considers both intermolecular interactions and tissue-specific expression levels of miRNA.

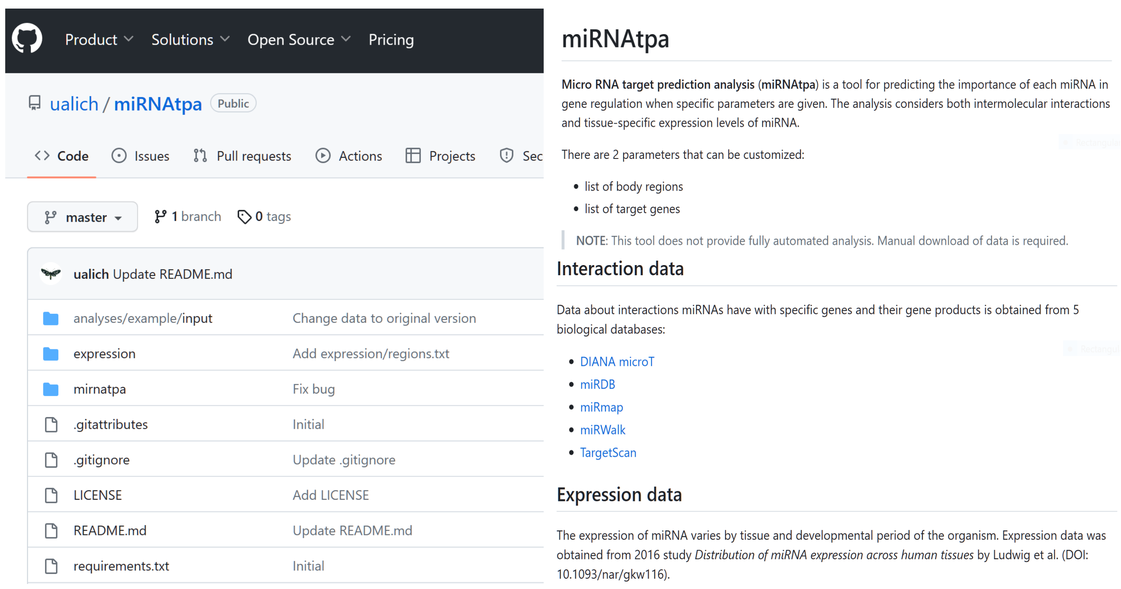

Figure 1. miRNA target prediction analysis tool – **miRNAtpa**.

Use case

We used five publicly available databases/web algorithms and six preselected candidate genes, which had emerged as promising candidates in the context of suicide research through previous research (*SLC6A4*, *HTR1A*, *BDNF*, *NR3C1*, *ZNF714*, and *NRIP3*) (Table 1). Based on prediction algorithm we have chosen miRNAs that are targeting regulation of the genes listed, and are at the same time being expressed in the brain (Table 2).

Expression of these miRNAs was tested in the brain tissue of 40 suicide completers and controls, and hsa-miR-4516 and hsa-miR-381-3p showed a trend for statistical significance (Figure 2).

Table 1. Number of predicted miRNAs involved in each target gene regulation.

Database	BDNF	HTR1A	SLC6A4	NR3C1	ZNF714	NRIP3
miRWalk	2353	1462	1942	2349	1858	1916
miRmap	979	48	1300	370	65	1202
TargetScan	775	153	1115	976	1506	875
DIANA microT-CDS	272	1	201	301	303	211
miRDB	191	21	161	472	269	163

Table 2. Final miRNA selection. The table contains separated values of target gene analysis and expression analysis, as well as combined values.

Selected miRNAs	Target interaction analysis	Expression analysis	Weighted sum
hsa-miR-4516	13.63	17.17	30.80
hsa-miR-3135b	16.75	13.57	30.33
hsa-miR-124-3p	11.55	17.87	29.43
hsa-miR-129-5p	20.89	8.47	29.35
hsa-miR-27b-3p	15.80	12.34	28.14
hsa-miR-381-3p	20.01	7.48	27.49
hsa-miR-4286	9.37	18.06	27.43

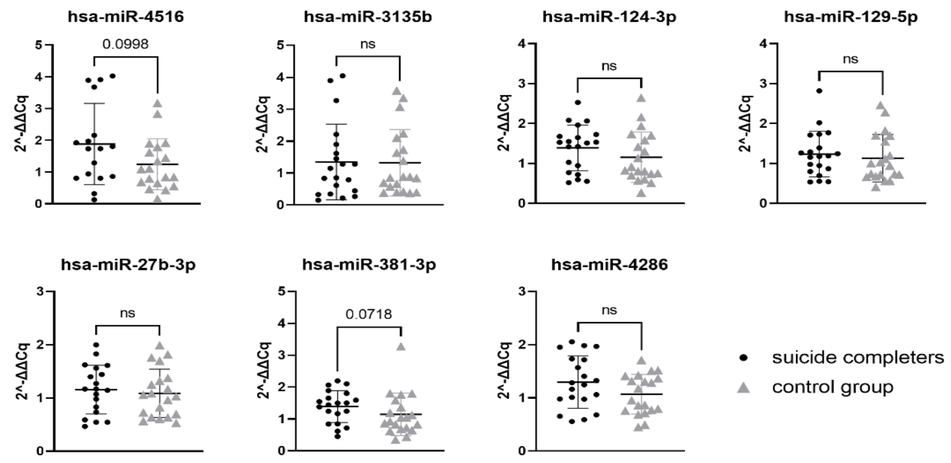

Figure 2. Relative gene expression of selected miRNAs in Brodmann area 0. Data are presented as a scatter plot, with each point being a measure of relative gene expression per subject for both studied groups. Horizontal lines represent mean value ± standard deviation. Data were analysed using the Mann–Whitney U test, with *p*-values below 0.05 as significant. Ns –not siginificant.

1. Videtič Paska A, et al. Suicide and Changes in Expression of Neuronal miRNA Predicted by an Algorithm Search through miRNA Databases. Genes (Basel). 2022;13(4).

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Contact

Alja Videtič Paska
Researcher/Center for Functional Genomics and Bio-Chips, Institute of Biochemistry and Molecular Genetics, Faculty of Medicine, University of Ljubljana, 1000 Ljubljana, Slovenia; ELIXIR Slovenia
Alja.videtic@mf.uni-lj.si

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